

Exponential Notation — Answers

Final answers with a one-line reason. For full methods, see the worked examples in the notes.

Objective

Q1 (b) **exponent.** In 5^4 , 5 is the base and 4 is the exponent (power).

Q2 (a) $2^3 \times 3^4$. $648 = 2 \times 2 \times 2 \times 3 \times 3 \times 3 \times 3$.

Q3 **16.** Even power of a negative base is positive: $(-2)^4 = 16$.

Q4 (b) **-16.** Without brackets, $-2^4 = -(2^4) = -16$.

Q5 **True.** Odd power of a negative base is negative: $(-5)^3 = -125$.

Q6 (a) **Both true, R explains A.** Each square is positive (9 and 25), giving 225.

Short answer

Q7 (i) 2^2a^2 (ii) $5^2 \times 7^3$.

Q8 $405 = 3^4 \times 5$; $540 = 2^2 \times 3^3 \times 5$.

Q9 **90000.** $3^2 \times 10^4 = 9 \times 10000$.

Long answer

Q10 (i) $648 = 2^3 \times 3^4$ (ii) $405 = 3^4 \times 5$ (iii) $540 = 2^2 \times 3^3 \times 5$.

Q11 (i) 225 (ii) 90000 (iii) -32000000. (Odd power of -2 is negative; even power of -10 is positive.)

Q12 $2^8 = 256$ cells. Doubling 8 times from 1 cell = 2^8 .

Total: 21 marks.